

Doan Phuong Anh-Pham

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Academic Portfolio

EDUCATION

Fulbright University Vietnam

2021 – 2025

Bachelor of Science with Honours in Computer Science
Minor: Human-Centered Engineering
GPA: 3.46 / 4.0

Tran Phu High School for the Gifted

2018 – 2021

Specialization in Biology
GPA: 9.2 / 10

- Scholarship for Excellent Specialized Students for 3 consecutive years
- Third Prize, Biology Specialized Category – City-Level Competition

RESEARCH EXPERIENCE

Agent Harness System for Medical Data

Jun 2026 – Present

Long-Chain Reasoning of Large Language Models for Diagnostic Knowledge Tracing and Image-Text Fusion

Advisor: Prof. Zengchang Qin

- Investigating long-chain reasoning strategies for multimodal medical AI, with a focus on diagnostic knowledge tracing and image-text fusion (MEDClip as the bridge)
- Reproduced and extended recent research by implementing and evaluating state-of-the-art medical reasoning and vision-language models
- Conducted experimental studies comparing LLM-based approaches with an agent-based architecture, contributing to a shift toward coordinated multi-agent clinical reasoning
- Developed experimental pipelines for benchmarking reasoning quality, multimodal integration, and agent collaboration across diagnostic tasks

Computational Neuroscience Research

Jul 2025 – Present

Structure-Aware Brain Connectivity Modeling from fMRI

Advisor: Dr. Duy Duong-Tran

- Leading independent hypergraph research while collaborating on CASTICaD, a project investigating brain connectivity and causal dynamics from fMRI BOLD time-series data
- Co-developed a spatiotemporal modeling framework for fMRI data; NeurIPS 2025 reviewer feedback helped refine mathematical foundations and experimental design
- Developing tensor-based hypergraph frameworks to model higher-order neural interactions beyond traditional pairwise connectivity
- Exploring hypergraph approaches for Transformer attention and topological descriptors to study synergy, redundancy, and multi-region coordination in brain dynamics
- Reviewing and integrating literature on Higher-Order Transformers, Tensor Attention Training, O-information, and hypergraph reconstruction from dynamics
- Code: [GitHub Repository](#)

Honors Capstone Project

Jun 2024 – Jun 2025

Designing a Photorealistic Simulation Platform for Robust Semantic Segmentation in Autonomous Vehicles

Advisor: Dr. Dang The-Huynh

- Developed a 3D photorealistic simulator using Houdini and Unreal Engine 5
- Generated approximately 8,000 synchronized RGB, semantic, and depth images via EasySynth
- Fine-tuned DeepLabV3 ResNet50 for 9-class traffic scene segmentation
- Evaluated performance using accuracy metrics, Dice coefficients, confusion matrices, and visual analysis
- [Thesis](#) · [GitHub Repository](#)

Queer Narrative Game Research

Jan 2024 – Jun 2024

Queer Women's Stories Through Interactive Gameplay

Advisor: Dr. Trung Thanh-Phan

- Co-developed a narrative-driven 2D RPG using Godot and GDScript
- Designed branching dialogue, relationship mechanics, and narrative systems
- Conducted mixed-methods user research with 100 participants and qualitative interviews
- [GitHub Repository](#)

INDUSTRY EXPERIENCE

AI Developer – VinAI in Action

Apr 2026 – Jun 2026

Vietnam

- Worked in the AI Infrastructure and Agentic AI track, developing production-ready AI systems and infrastructure
- Built ETL pipelines and data ingestion workflows to support large-scale AI training and inference
- Contributed to the design and implementation of data lake architecture for efficient data storage, processing, and retrieval
- Developed agentic AI workflows integrating LLMs, external tools, and autonomous task execution
- Collaborated on AI infrastructure, deployment pipelines, and system integration for scalable machine learning applications

AI Engineer – Quest Ventures

Jun 2025 – Present

Singapore

- Maintained and extended internal systems using PHP and JavaScript
- Built automation tools and backend services for internal workflows
- Deployed and administered an internal Mattermost chat server
- Managed servers, APIs, performance optimization, and documentation

AI & Software Engineering Intern – ForthAI

Jun 2024 – Sep 2024

Singapore

- Implemented AI agent architectures using AutoGen, LangChain, LangGraph, and crewAI
- Built plan-action and management agents for autonomous execution
- Worked with Next.js, GCP, and Cloud Build pipelines in a startup environment

Quality Control & Assurance Collaborator – Garena Vietnam

Apr 2024

- Tested gameplay, UI/UX, and graphics using Garena's internal engine
- Coordinated with Vietnamese and Chinese teams to report and resolve issues

AWARDS & HONORS

2nd Runner-Up, SheCodes Hackathon

2023

National hackathon recognizing innovative technical solutions by women in tech

2nd Runner-Up, Youth Digital Challenge Competition

2024

Hackathon focused on cybersecurity, digital connection, and secure technologies

REFERENCES

Dr. Dang The-Huynh

Professor of Computer Science, Fulbright University Vietnam

Dr. Duy Duong-Tran

Tenure-track Assistant Professor in the Department of Mathematics at the United States Naval Academy (USNA)